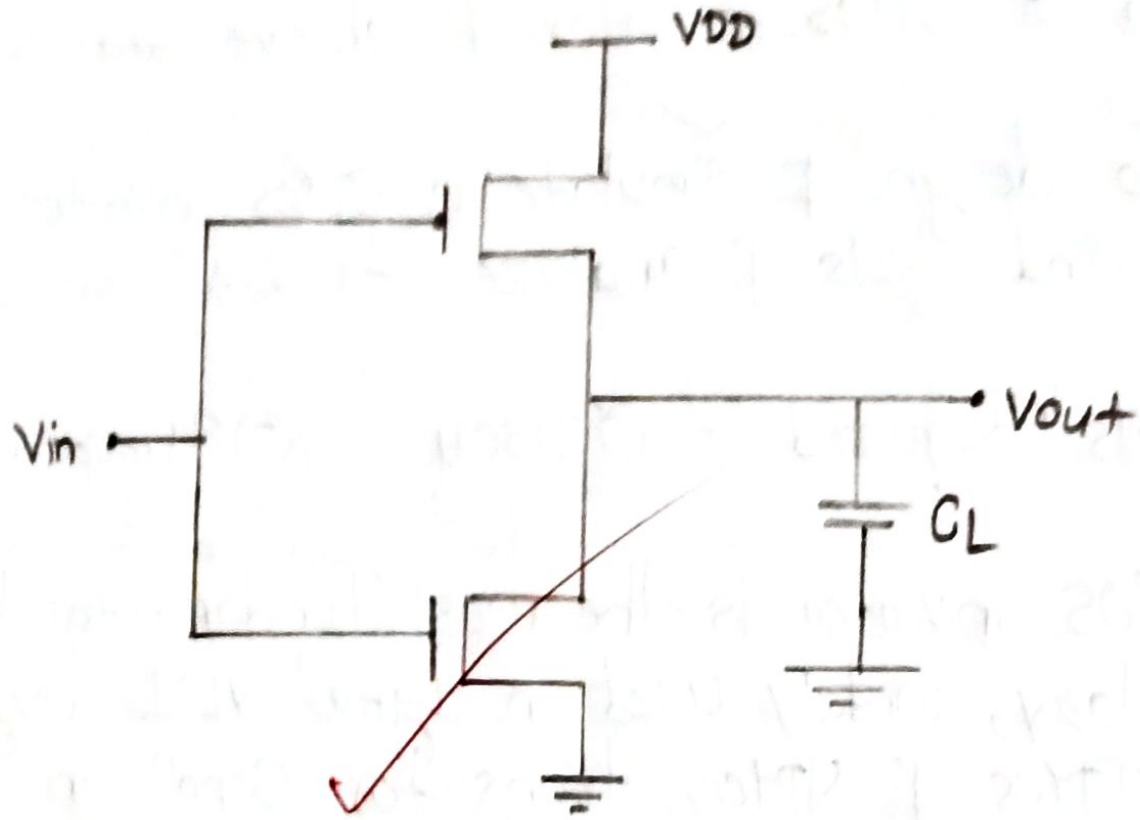
Truth Table

| A | O/P |
|---|---|
| 0 | 1 |
| 1 | 0 |




Schematic diagram of CMOS inverter

Truth Table :-

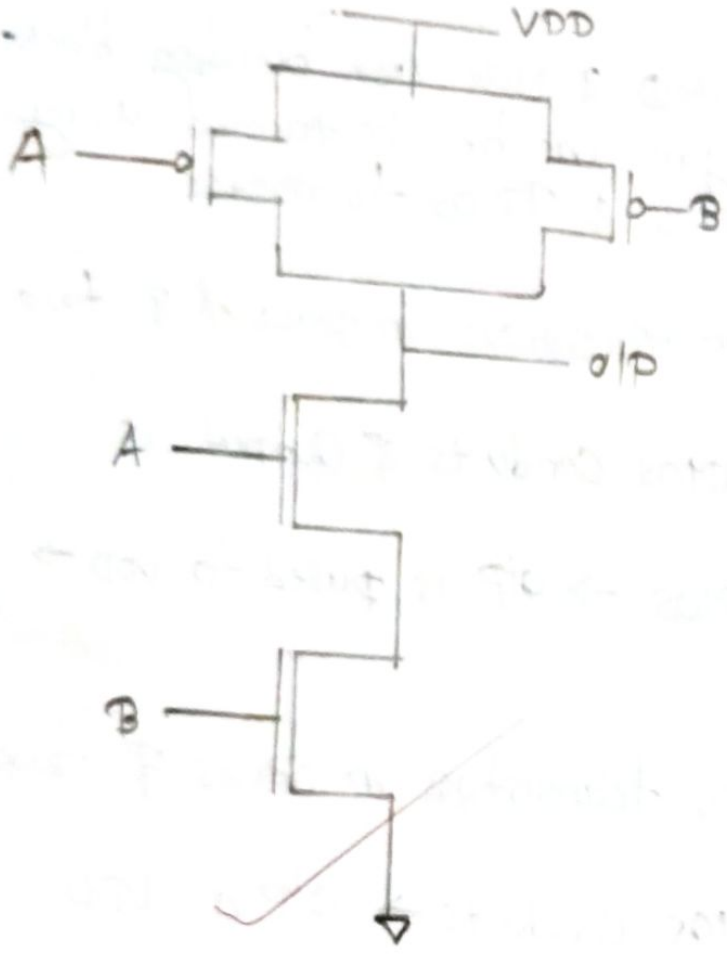
NAND Gate

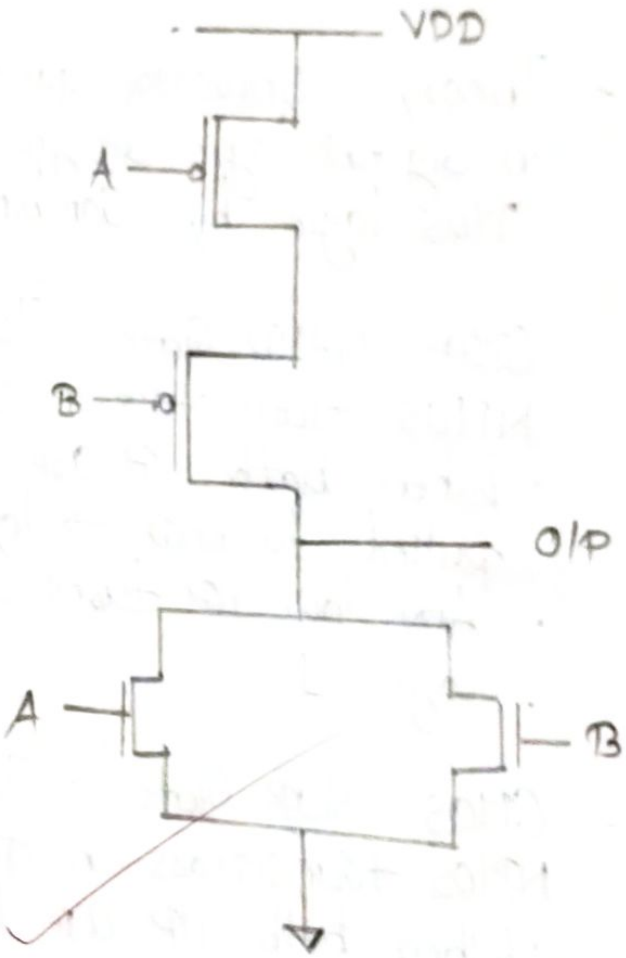
| A | B | Y = (AB) |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

NOR Gate

| A | B | Y = (A+B) |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 0 |

Schematic Diagram :-



Truth Table :-

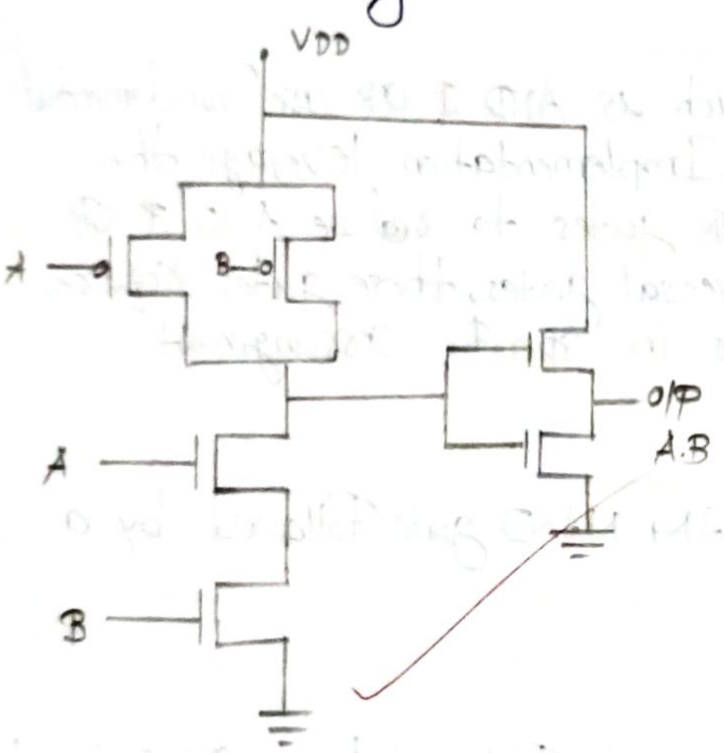
AND gate

| A | B | o/p A B |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

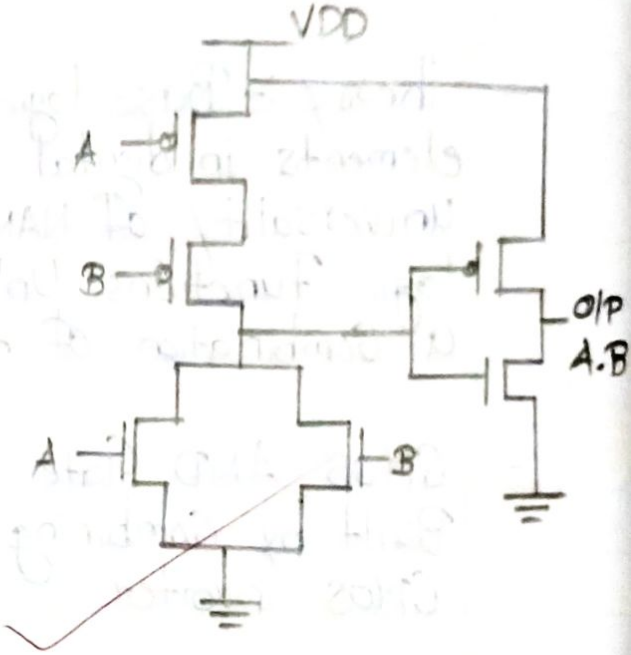
OR gate

| A | B | o/p A+B |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 1 |

Schematic Diagram :-




AND Gate



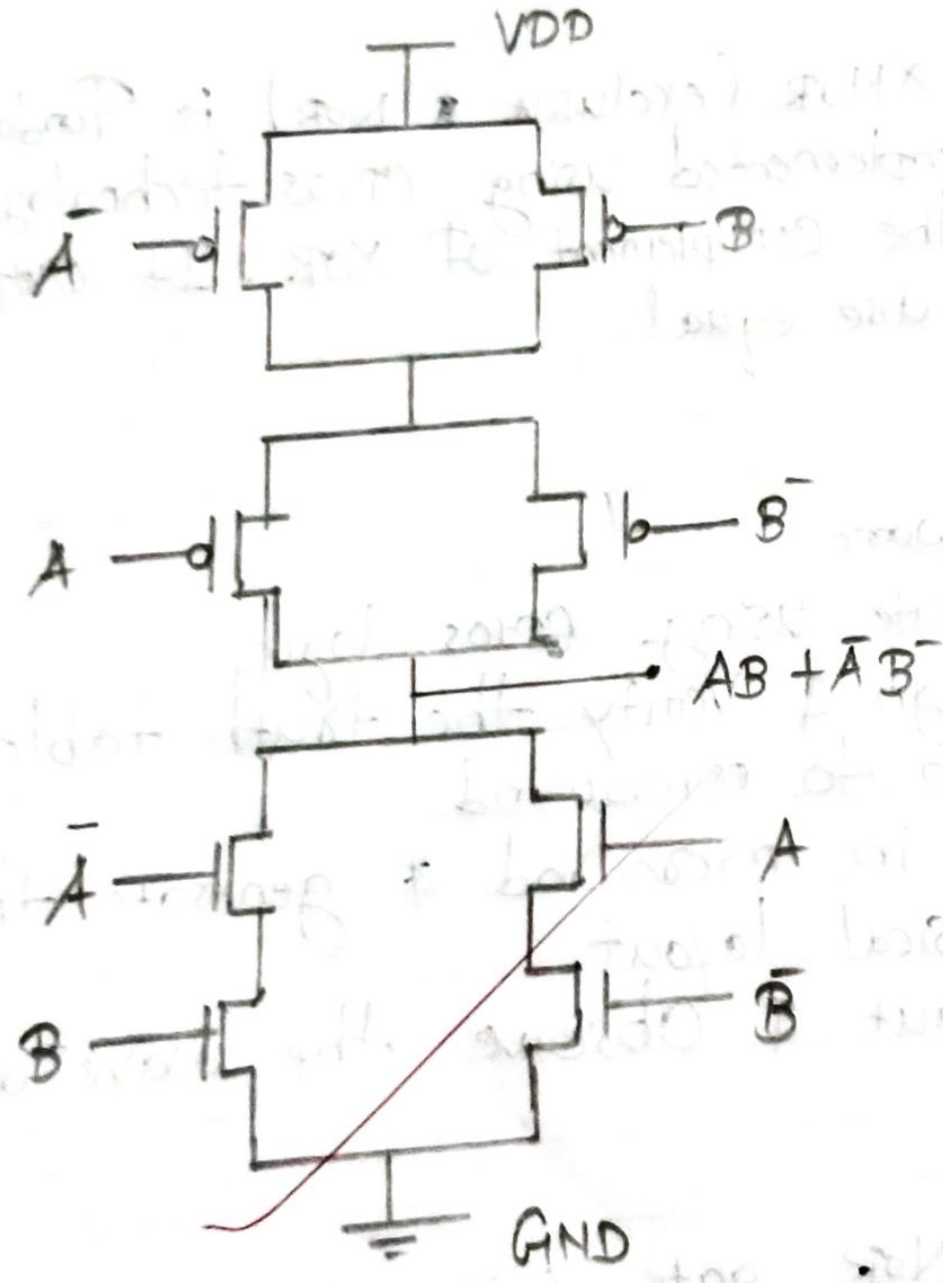

OR Gate

Truth table XNOR

| A | B | O/P |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

$Y = A \oplus B = A\bar{B} + \bar{A}B$

Schematic Diagram:-

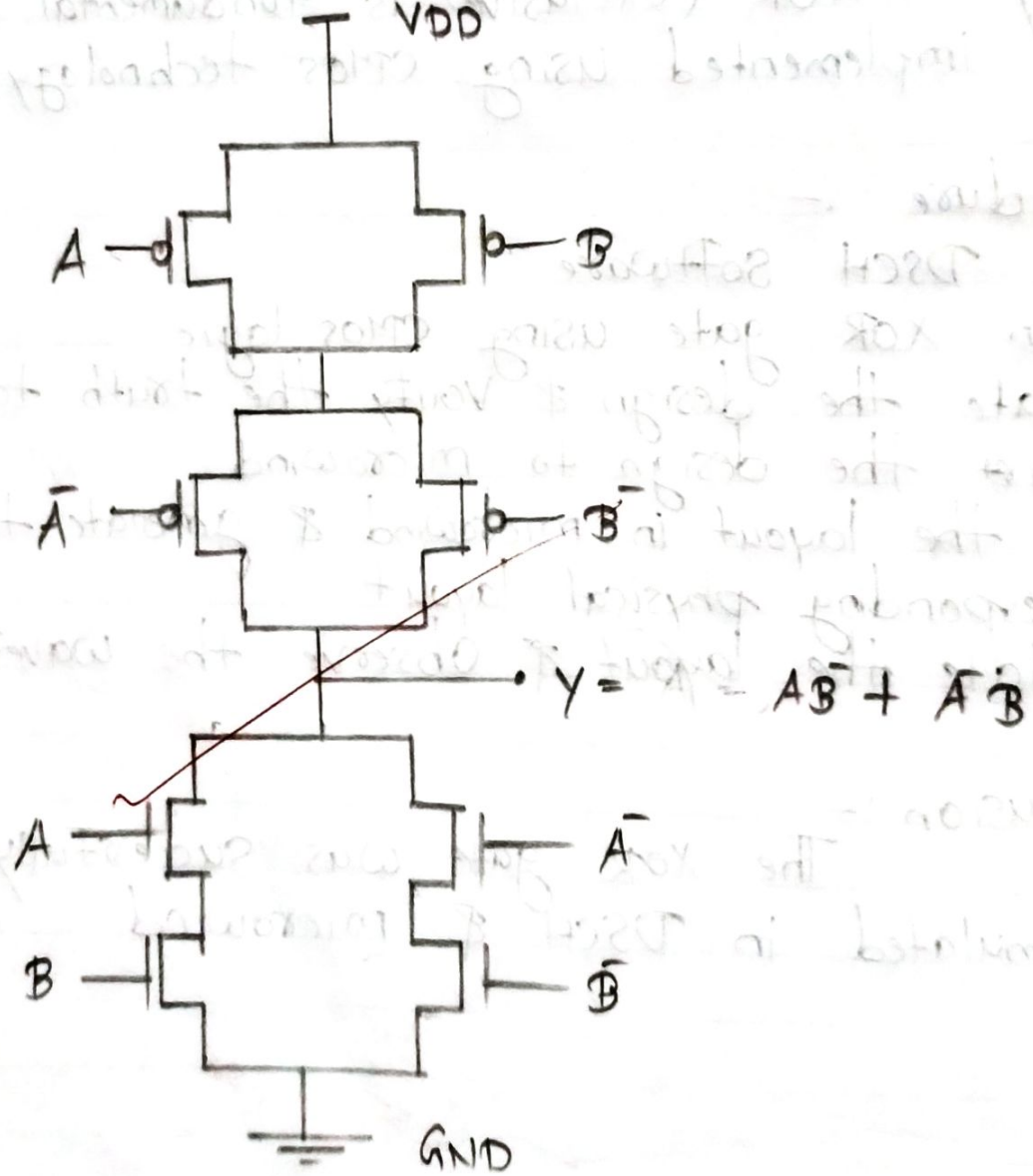


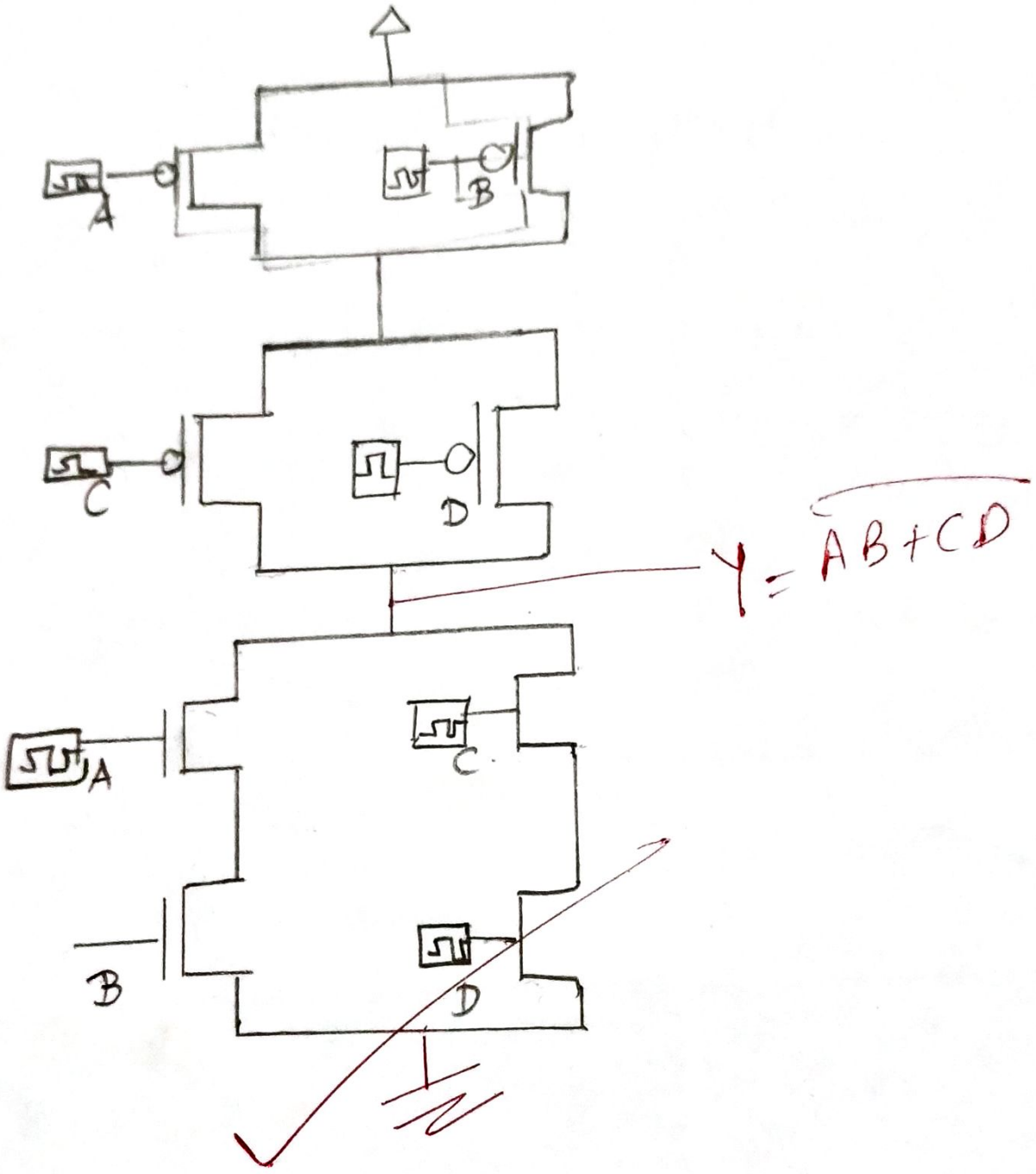
# Truth Table :- XOR

| A | B | o/p |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

$Y = A \oplus B = A\bar{B} + \bar{A}B$

## Schematic Diagram :-






Boolean expression

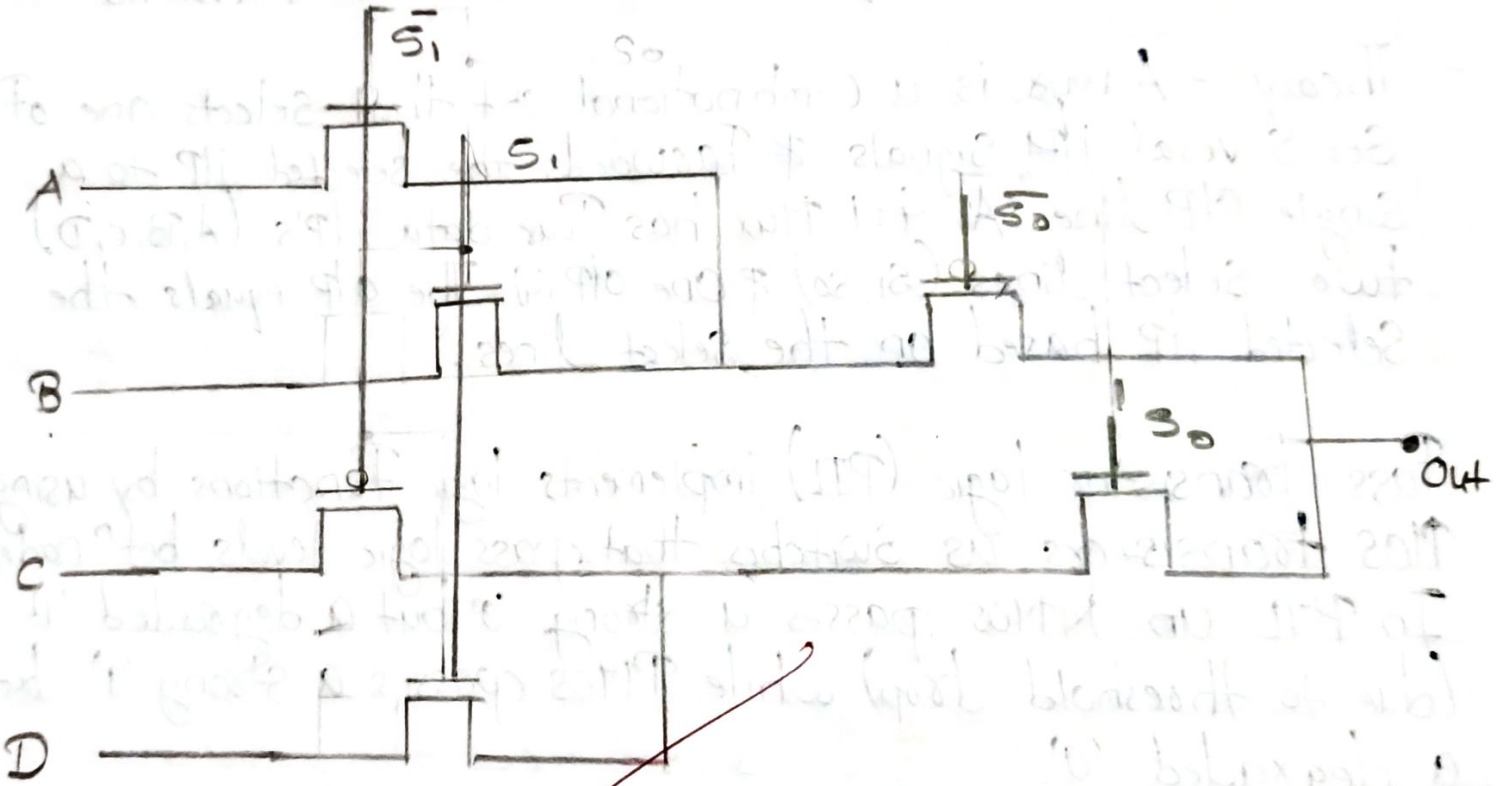
# Truth Table

| A | B | C | D | AB | CD | O/P (AB + CD) |
|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | 1 |
| 0 | 0 | 0 | 1 | 0 | 0 | 1 |
| 0 | 0 | 1 | 0 | 0 | 0 | 1 |
| 0 | 0 | 1 | 1 | 0 | 1 | 0 |
| 0 | 1 | 0 | 0 | 0 | 0 | 1 |
| 0 | 1 | 0 | 1 | 0 | 0 | 1 |
| 0 | 1 | 1 | 0 | 0 | 0 | 1 |
| 0 | 1 | 1 | 1 | 0 | 1 | 0 |
| 1 | 0 | 0 | 0 | 0 | 0 | 1 |
| 1 | 0 | 0 | 1 | 0 | 0 | 1 |
| 1 | 0 | 1 | 0 | 0 | 0 | 1 |
| 1 | 0 | 1 | 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 0 | 1 | 0 | 0 |
| 1 | 1 | 0 | 1 | 1 | 0 | 0 |
| 1 | 1 | 1 | 0 | 1 | 0 | 0 |
| 1 | 1 | 1 | 1 | 1 | 1 | 0 |

## Truth Table

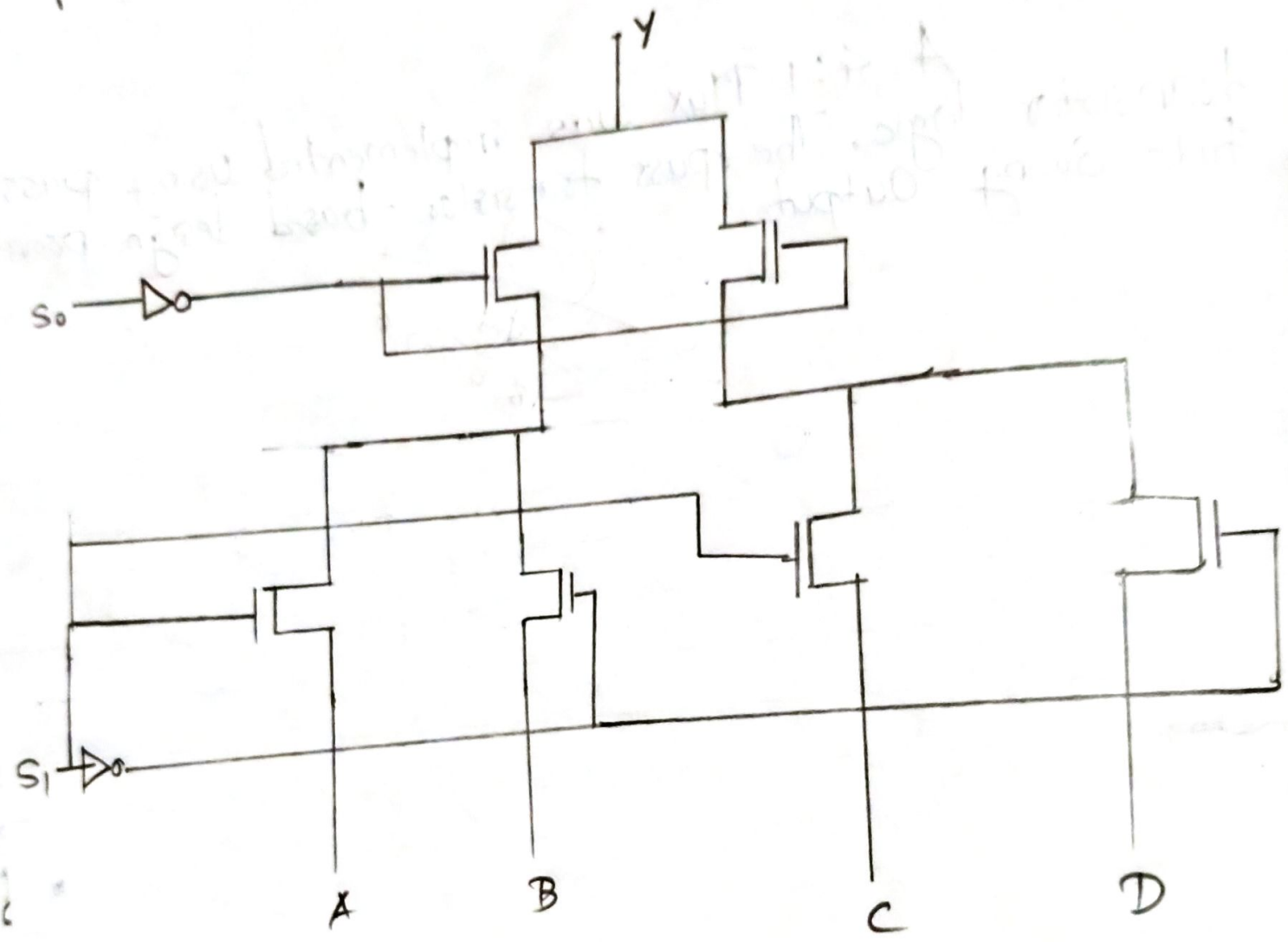
| $S_0$ | $S_1$ | A | B | C | D | Y |
|---|---|---|---|---|---|---|
| 0 | 0 | A | | | | A |
| 0 | 1 | | B | | | B |
| 1 | 0 | | | C | | C |
| 1 | 1 | | | | D | D |

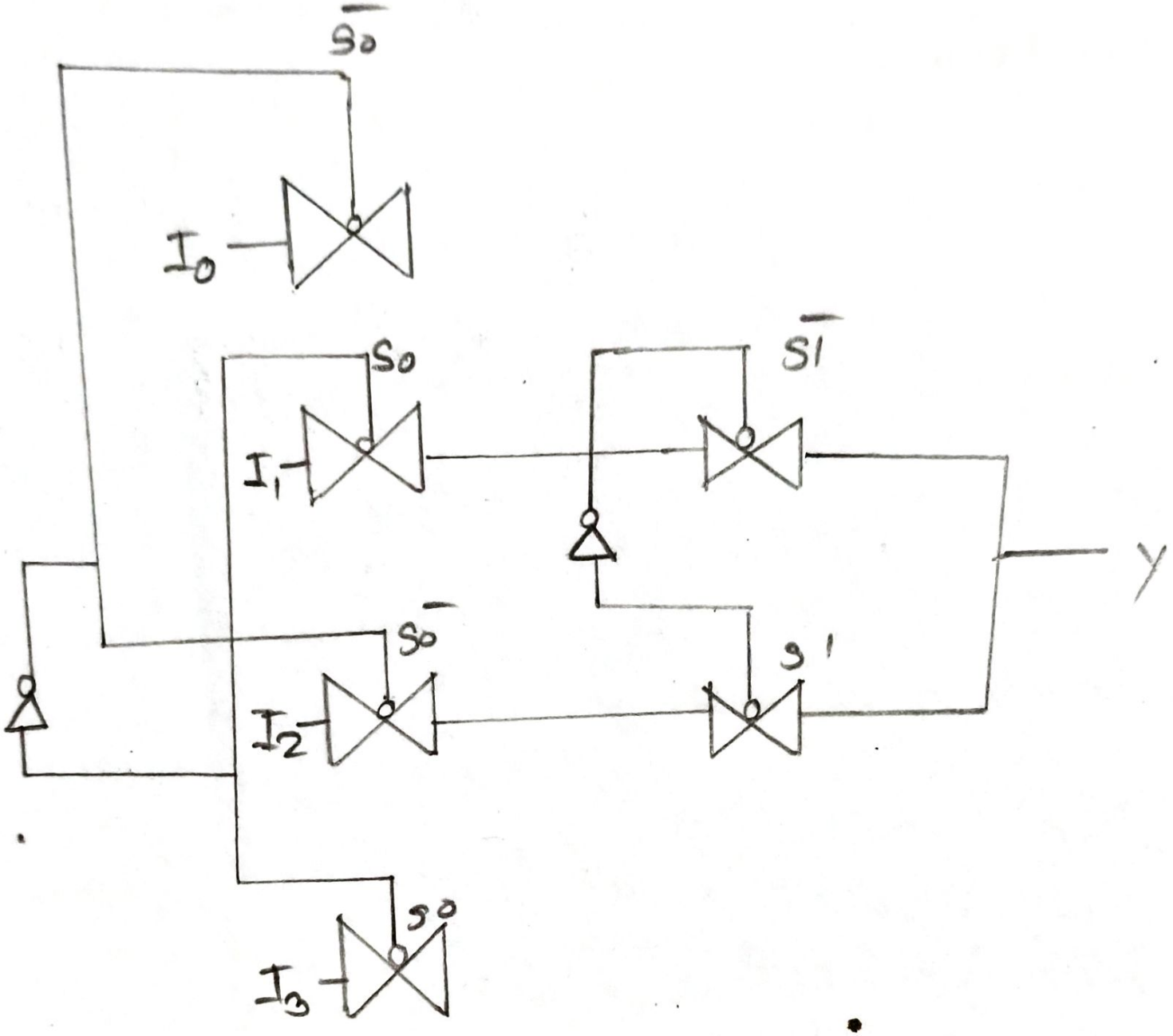
## Schematic Diagram -



Truth Table

4:1 Mux using Pass Transistor logic

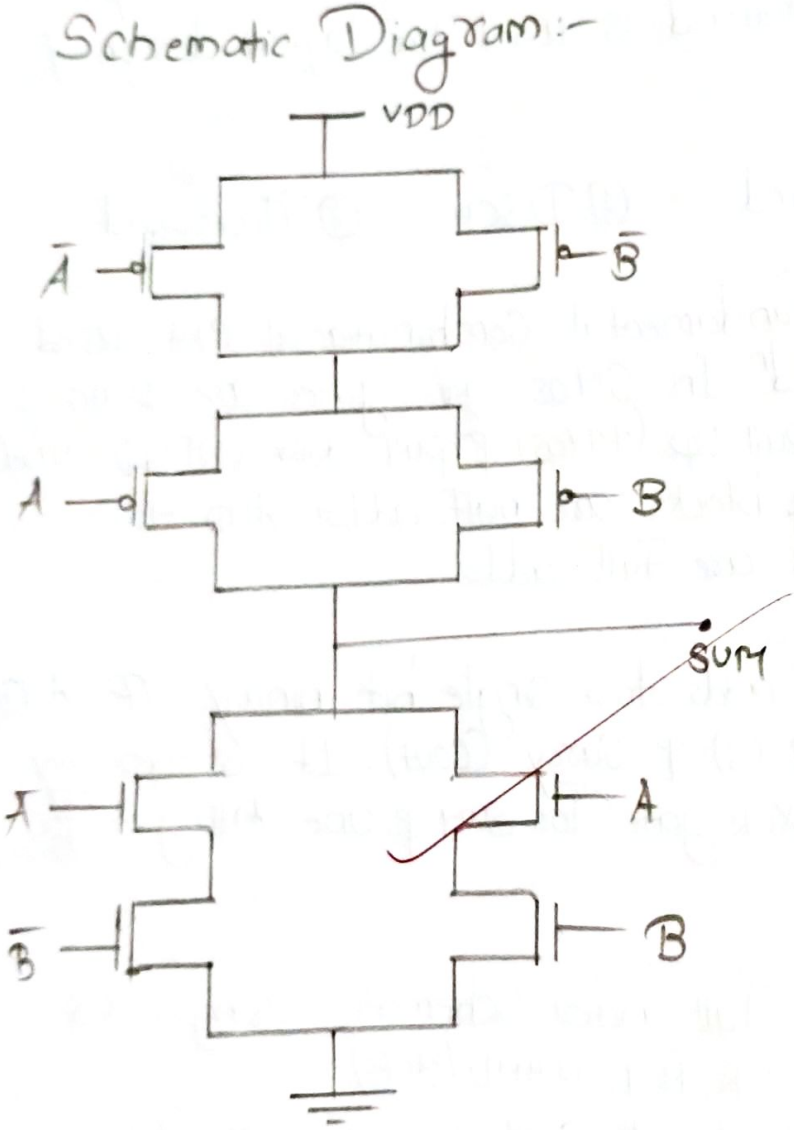


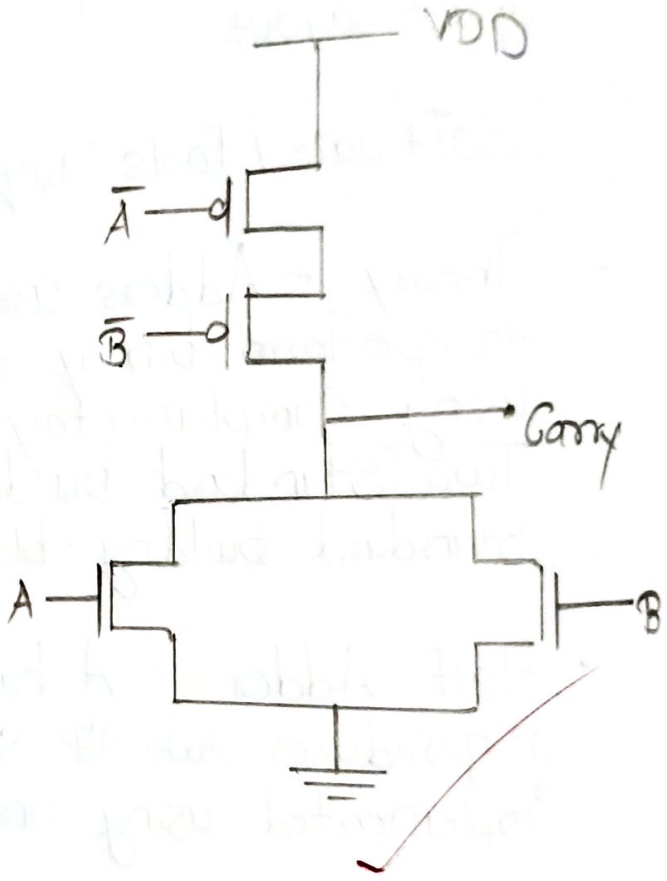

$\overline{S_0}$
$I_0$
$S_0$
$I_1$
$\overline{S1}$
Y
$\overline{S_0}$
$I_2$
S 1
$I_3$
S 0

Truth table :-

| A | B | Sum | Cout |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 1 | 1 | 0 |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 1 |

Schematic Diagram :-



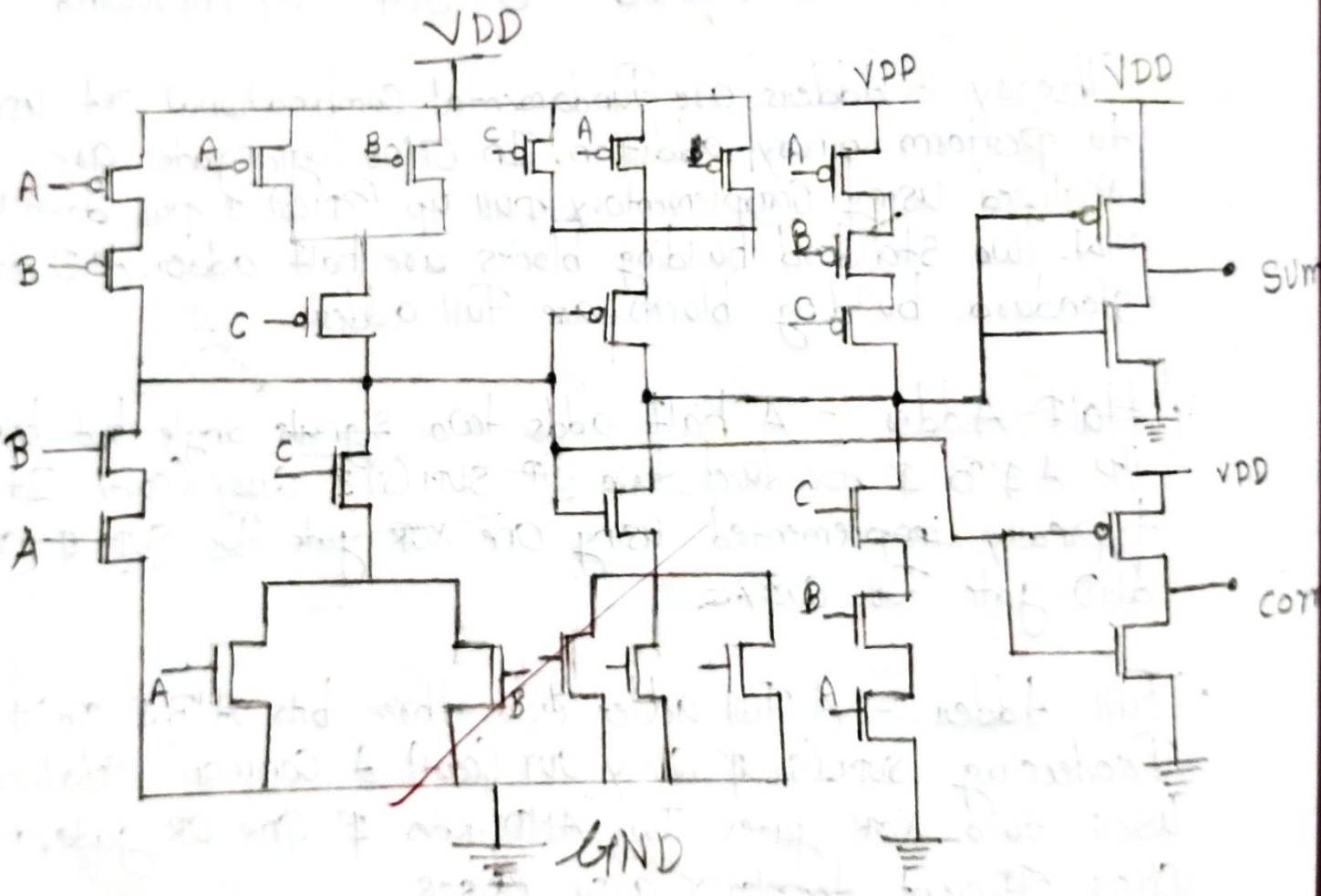




Half Adder

# Full Adder (Truth Table):-

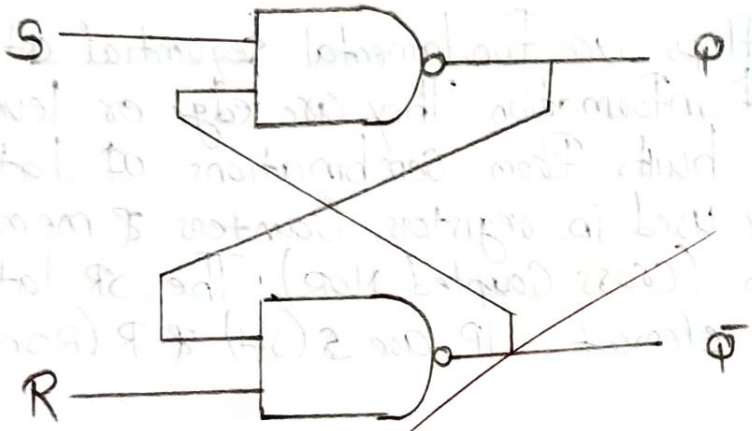
| A | B | Cin | S (sum) | Cout (carry) |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 | 0 |
| 0 | 1 | 0 | 1 | 0 |
| 0 | 1 | 1 | 0 | 1 |
| 1 | 0 | 0 | 1 | 0 |
| 1 | 0 | 1 | 0 | 1 |
| 1 | 1 | 0 | 0 | 1 |
| 1 | 1 | 1 | 1 | 1 |

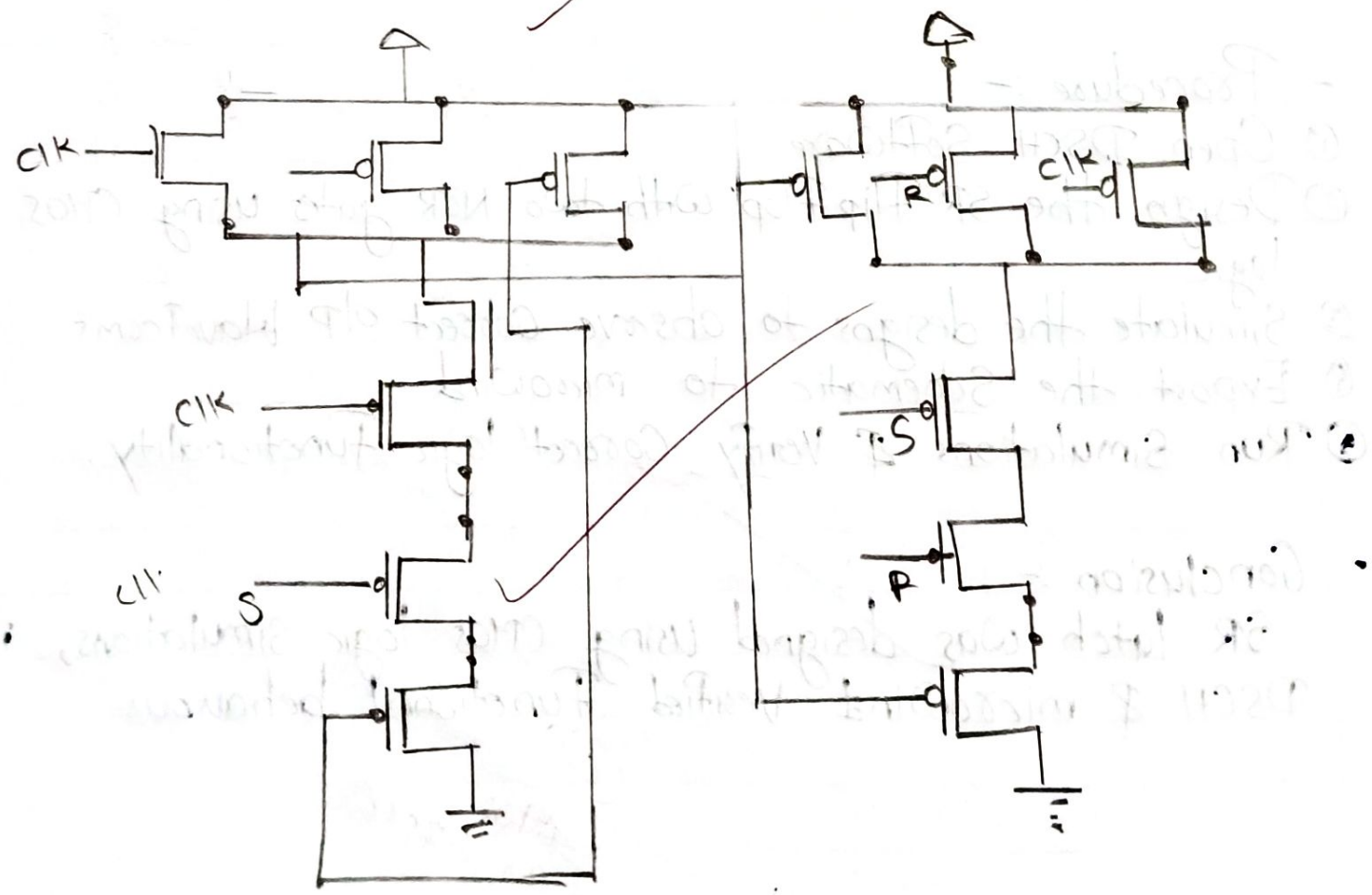
## Schematic Diagram :-

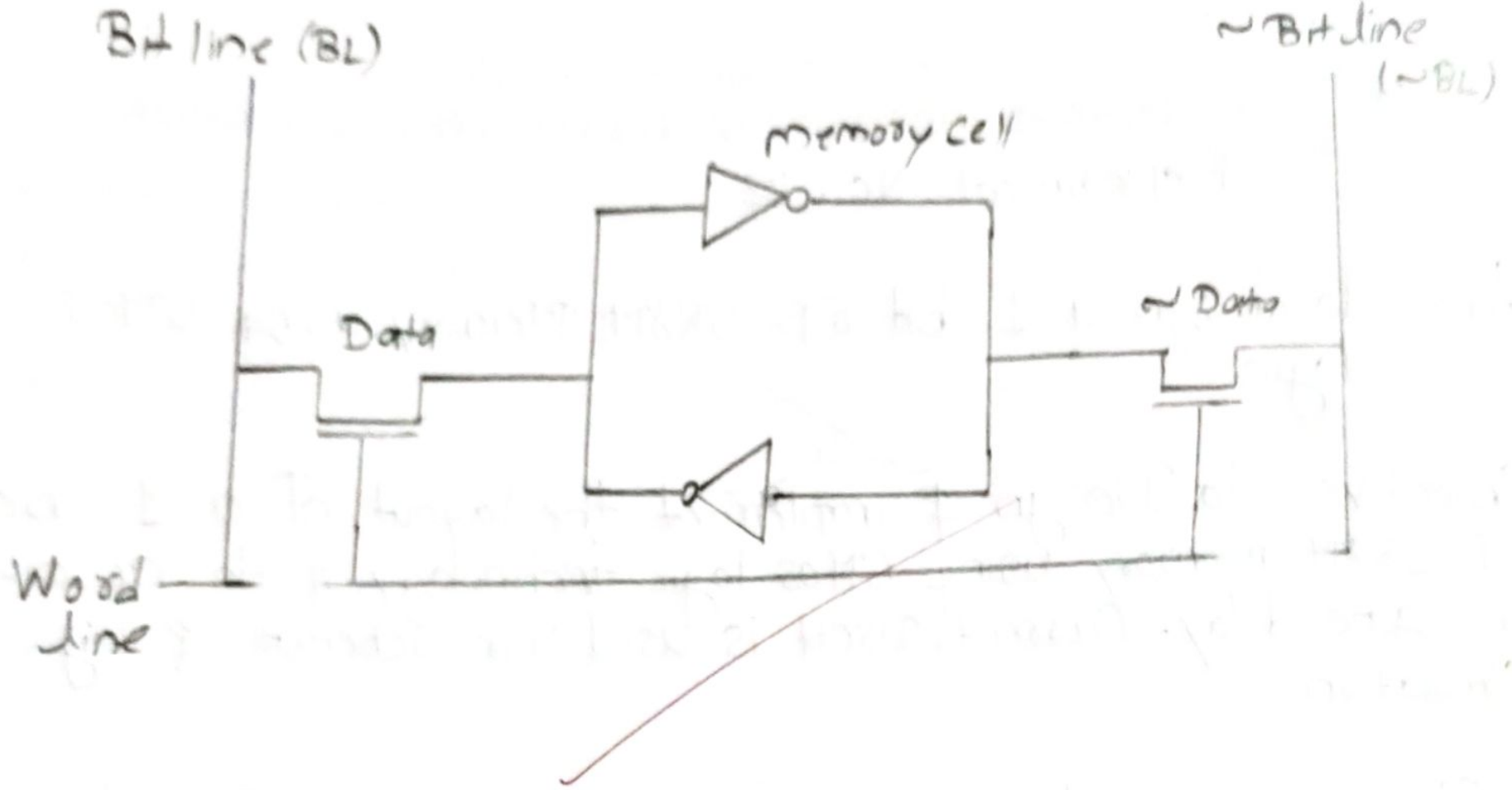


# Truth Table

| S | R | Q (next) | Remark |
|---|---|---|---|
| 0 | 0 | Q (prev) | No Change |
| 0 | 1 | 0 | Reset |
| 1 | 0 | 1 | Set |
| 1 | 1 | undefined | Invalid |




Bit line (BL)
~Bit line
(~BL)
memory cell
Data
~Data
Word
line